Mohammod Ashikur Rahman

ashikbuilds | Mohammod Ashikur Rahman | Google Scholar
mdashikurrahman2418@gmail.com | 01586357555

Summary

Computer Science graduate and Research Assistant with expertise in Machine Learning, Deep Learning, Natural Language Processing, and Explainable AI. Experienced in developing AI-driven solutions for healthcare analytics, multilingual hate speech detection, and predictive modeling. Published author at ICCIT 2025 with ongoing research in graph neural networks, transformer-based learning, and healthcare AI.

Work Experience

Research Assistant, Al Hikma Research Lab May 2026 – Present

Conducting research in Machine Learning, Deep Learning, Natural Language Processing (NLP), and Computer Vision. Developing AI models, performing literature reviews, preparing datasets, and contributing to publication-oriented research in healthcare AI and explainable machine learning.

Projects

Transformer Ensemble for Hate Speech Detection GitHub

Developed a multilingual NLP system using BERT, LSTM, RoBERTa, and DeBERTa for hate speech detection. Implemented ensemble learning techniques to improve performance across datasets.

DR-Detection-using-ViT GitHub

Developed a deep learning-based system for diabetic retinopathy detection using Vision Transformer (ViT) on retinal fundus images. Implemented preprocessing, model training, and evaluation pipeline with performance metrics analysis.

AccidentNet-XAI GitHub

Developed an explainable ensemble learning framework for road accident severity prediction using stacking models. Applied feature selection (Sequential Backward Elimination), class balancing, and interpretability techniques including SHAP and permutation importance to achieve high-performance multiclass classification.

Agentic AI Framework using LangGraph GitHub

Built a modular agentic AI system using LangGraph enabling autonomous decision-making, multi-step reasoning, and tool integration for scalable AI workflows.

Book-Vibee (Frontend Web Project) GitHub

Developed a responsive frontend web application for book-related content and user interaction, focusing on intuitive UI design and seamless user experience.

Social Media Platform (Java) GitHub

Designed and implemented a Java-based social networking application with core features such as user interaction, content sharing, and basic system functionality.

Skills

- **Programming:** Python, C, C++, Java, SQL
- **Machine Learning & AI:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Transformer Models, Large Language Models (LLMs), Explainable AI (XAI), En-

semble Learning, Agentic AI (LangGraph)

- **Libraries & Frameworks:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face Transformers
- **Tools & Technologies:** Git, GitHub, Linux, REST APIs, Firebase, MongoDB, MySQL, SQLite
- **Concepts:** Data Structures & Algorithms, Data Preprocessing, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Model Deployment
- **Other:** Competitive Programming (Codeforces), Problem Solving, Bash Scripting, Research Writing

Publications

Published Paper

Interpretable Machine Learning for Proactive Diabetes Screening IEEE Xplore

- Published in ICCIT 2025. Developed an interpretable stacking ensemble framework for proactive diabetes prediction using BRFSS healthcare data with SHAP and LIME analysis.

Under Review

Explainable Deep Graph Neural Networks for Multi-Stage Hepatitis C Classification GitHub

Submitted to *Healthcare Technology Letters*. Developed an explainable GNN-based framework for multi-stage Hepatitis C classification using patient similarity modeling.

Transformer-Based Multilingual Hate Speech Detection GitHub

Under review in *Machine Learning with Applications* (Elsevier, Q1). Developed transformer-based ensemble models for multilingual and cross-lingual hate speech detection tasks.

Achievements

- Secured **4th Place** among the **Top 9 Selected Candidates** in the **Research-Oriented Competitive Coding Event 2026** organized by BUBT Research Graduate School (BRGS). Result
- Participated in BUBT Artificial Intelligence Conquest (BAIC) 2025, a national-level AI competition, demonstrating skills in problem-solving, machine learning, and analytical thinking. Certificate
- Solved 100+ problems on Codeforces, demonstrating strong problem-solving skills and understanding of algorithms and data structures. Profile
- Completed Machine Learning Specialization by DeepLearning.AI (Stanford University, Coursera, 2026), covering Supervised Learning, Advanced Learning Algorithms, and Unsupervised Learning with Recommender Systems and Reinforcement Learning. Achieved up to 100% in coursework. Certificate
- Participated in BUBT Intra University Capture The Flag (CTF) 2026 competition, demonstrating strong problem-solving, cybersecurity, and analytical skills. Certificate

Education

Bachelor of Science in Computer Science and Engineering, Bangladesh University of Business and Technology (BUBT) (CGPA: 3.70/4.00) 2022 – 2026